HEX2TAPE_LGP30 Manual

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Applicable to: HEX2TAPE_LGP30 Version 1.0

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Disclaimer

This manual and the HEX2TAPE_LGP30 program which it describes are free (as in beer) and free (as in thought). Do with them as you will, provided that you accept all of the consequences.

Introduction

HEX2TAPE_LGP30 converts Intel-hex-formatted binaries produced by XASM_LGP30 to v-formatted binaries for the LGP-30. The start address (if any) in the input file is carried across to a Stop and Transfer record in the output file.

Support

The program has been written to meet the author's personal requirements. It has been shared for what it is worth, but major changes are not envisaged. If you wish, report installation problems, bugs and feature requests using the relevant GitHub features. This is the only way you will get a response, which in any case is not guaranteed.

Installation

The author has minimal packaging and installation skills. The following should work:

1. Clone the GitHub repository.

git clone https://github.com/etherbrat/hex2tape_lgp30

2. This will create a structure which looks like:

Structure	Actual
project_dir	hex2tape_lgp30
dist_dir (distribution)	dist
docs_dir (documentation)	docs
src_dir	src
modules_dir (python files)	hex2tape_lgp30
tests_dir (tests)	tests
venv_dir (interpreter, libraries)	venv
etc.	etc.

3. The .exe file in the dist directory should run under Windows.

To build from source:

4. In the project directory, type:

python -m venv venv (create a virtual environment)
.\venv\scripts\activate (activate the virtual environment)
pip install -r requirements.txt (install the dependencies)
pip install -e . (the dot is a literal - current directory)

5. In the project directory, type:

pip install pyinstaller
pyinstaller --onefile .\src\hex2tape_lgp30\hex2tape.py (this will generate a .exe file in the dist directory)

The generated .exe file should run under Windows. If you want, you can copy the .exe file to a Windows Program Files subdirectory and create shortcut to a command window which has the .exe file on its path.

Invocation

In the command window, type:

hex2tape <path_to_hex_file>

If the input file is called foobar.baz, the output file will be called foobar.baz.lgp30_tape.

Appendix 1 – hex2tape Session (helloworld)

```
>assembler helloworld.asm -o helloworld.hex -l helloworld.lst
XASM_LGP30 Version 1.0
Listing file: helloworld.lst
Source file: helloworld.asm
End of pass 1 - 0 errors
End of pass 2 - 0 errors
Object file: helloworld.hex

>hex2tape helloworld.hex
HEX2TAPE_LGP30 version 1.0
Drum: Segment byte range: 0 - 488
Address: 0 <-- 65564
[etc.]
[etc.]
Address: 60 <-- 4
Drum: Program start address: 0

>simulator primary_boot.hex
[ed. but note that the primary boot loader is not used in this example]
SIM_LGP30 version 1.0
Reading hex file: primary_boot.hex
Address: 4032 <-- 524288
[etc.]
[etc.]
Address: 4037 <-- 0
Start address in hex file: 4032
--> tape2drum helloworld.hex.lgp30_tape
Reading tape file: helloworld.hex.lgp30_tape
Address: 0 <-- 65564
[etc.]
[etc.]
Address: 60 <-- 4
Start address in tape file: 0
--> run
Expecting keyboard input
Hello World
Good night and good luck
--> exit
>
```

Appendix 2 – hex2tape Session (blackjack)

```
>simulator primary_boot.hex
[ed. but note that the primary boot loader is not used in this example]
SIM_LGP30 version 1.0
Reading hex file: primary_boot.hex
Address: 4032 <-- 524288
[etc.]
[etc.]
Address: 4037 <-- 0
Start address in hex file: 4032
--> tape2drum bkjck.txt
Reading tape file: bkjck.txt
Address: 2880 <-- 864140
[etc.]
[etc.]
Address: 3519 <-- 0
[ed. the original blackjack tape does not have a start address - add one]
--> sta 2880
Start address: 2880
--> drum2hex bkjck.hex
--> exit

>hex2tape bkjck.hex
HEX2TAPE_LGP30 version 1.0
Drum: Segment byte range: 23040 - 28160
Address: 2880 <-- 864140
[etc.]
[etc.]
Address: 3519 <-- 0
Drum: Segment byte range: 32256 - 32304
Address: 4032 <-- 524288
[etc.]
[etc.]
Address: 4037 <-- 0
Drum: Program start address: 2880

>simulator primary_boot.hex
[ed. but note that the primary boot loader is not used in this example]
SIM_LGP30 version 1.0
Reading hex file: primary_boot.hex
Address: 4032 <-- 524288
[etc.]
[etc.]
Address: 4037 <-- 0
Start address in hex file: 4032
--> tape2drum bkjck.hex.lgp30_tape
Reading tape file: bkjck.hex.lgp30_tape
Address: 2880 <-- 864140
[etc.]
[etc.]
Address: 4037 <-- 0
Start address in tape file: 2880
--> blackjack on
Blackjack mode: On
--> tabs 15 30 45
Tab stops: 15 30 45
--> run
Expecting keyboard input
None A/ac in c instruction at 2880 - substituting zero

              5 - c          7 - s
              4 - s
Card?
Input terminated by non-Friden character
--> exit
>
```